A general framework for using two-stage individual participant data meta-analysis to predict individual treatment effects

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Background

- **Individualized Treatment Effect (ITE)**: relative benefit of an intervention
- **Individual participant data meta-analysis (IPD-MA)**: One-stage models are conceptually easier and widely used. A key application of IPD-MA is to build predictive models for ITE.
- With data sharing constraints, a **two-stage approach** is required

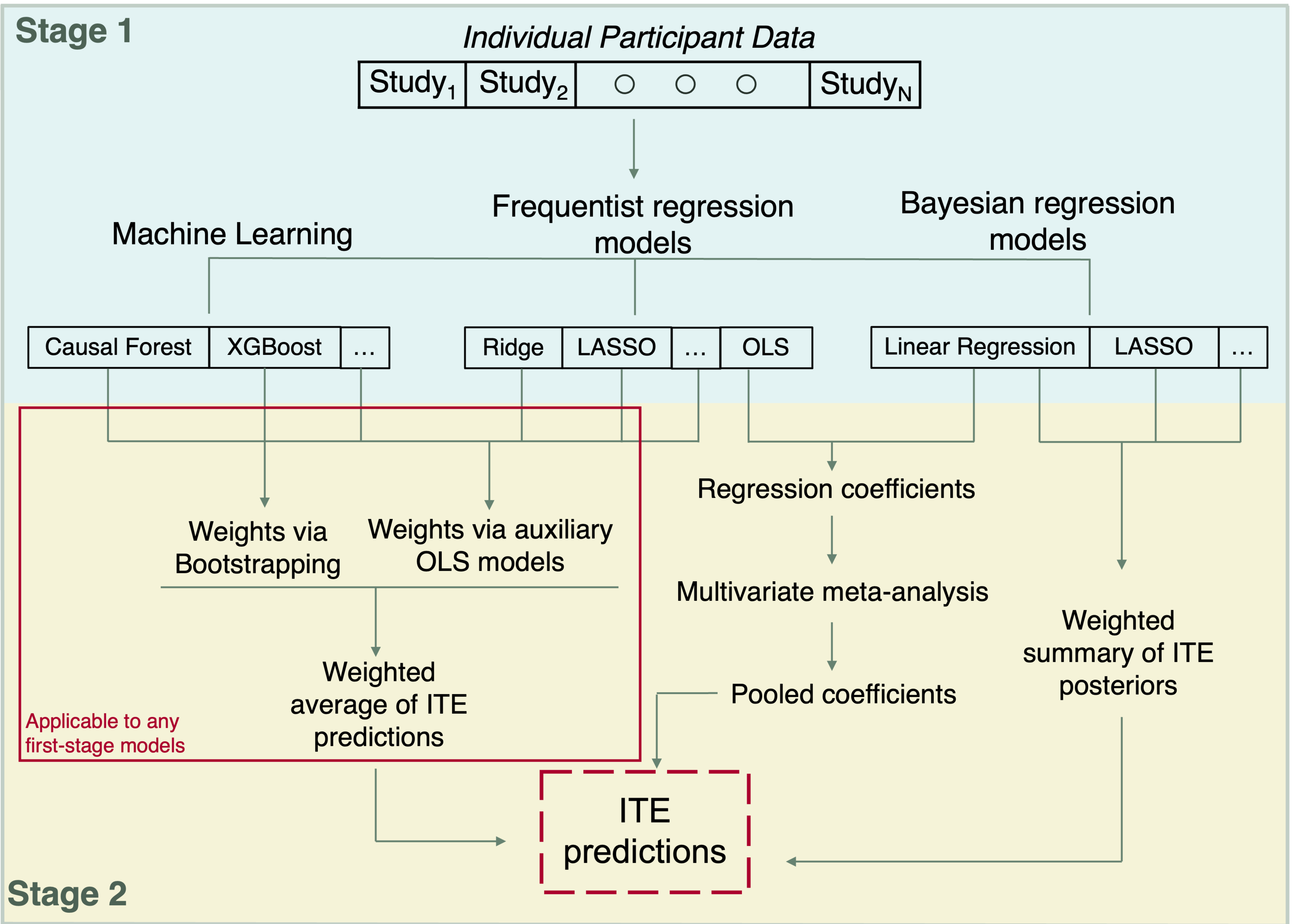
Aims

- Present a **general** two-stage IPD-MA **framework** to predict ITE
  - Flexibility – supports any type of models within trial to predict ITE
  - Adaptability – different first-stage models can be used across trials
- **Evaluation**: conduct a simulation study to assess the framework across scenarios
- **Simulation setup**: data generation mechanisms varying the following:
  - study-specific sample size (n)
  - linear or complex treatment-covariate interactions
  - treatment effect heterogeneity (τ^2)
  - effect modification (γ)

Methods: two-stage framework to predict ITE from IPD-MA

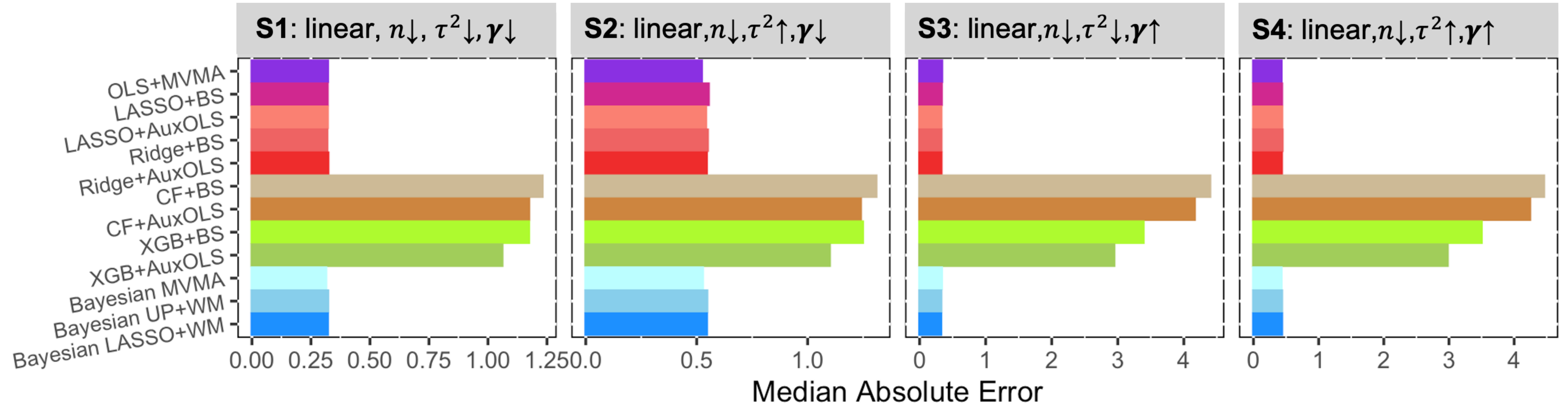
First stage
fit study-specific models to either predict ITE or estimate regression coefficients for the main treatment effect and effect modifiers

Second stage
pool study-specific results from first stage with a weighted average of predictions or meta-analysis of coefficients to get ITE predictions

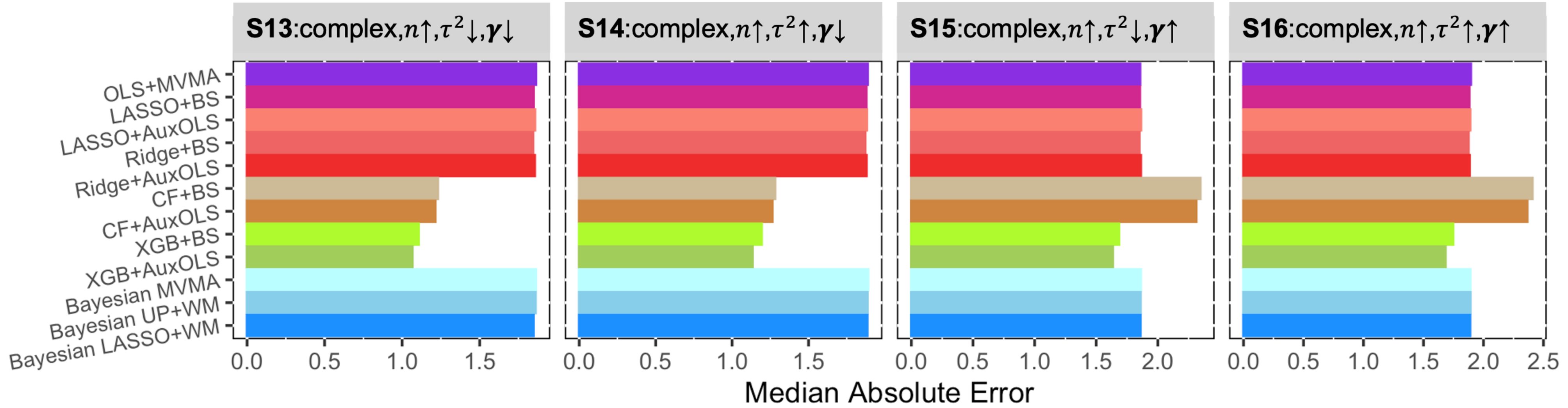


Results and key findings

- **Performance** varied across simulated scenarios
- **DGMs with linear interactions -** regression models used in the first stage outperformed machine learning approaches



- **DGMs with complex interactions -** machine learning performed best in large samples



- **Overall**, OLS inverse-variance weighting appeared to be a better pooling strategy than bootstrap.

Conclusions

- **IPD-MA** can support **clinical decision making** for **personalized treatment choice**
- **Performance** was highly dependent from DGMs
- However, DGMs is not known in real world data
- Model selection should be based on **proper validation**

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